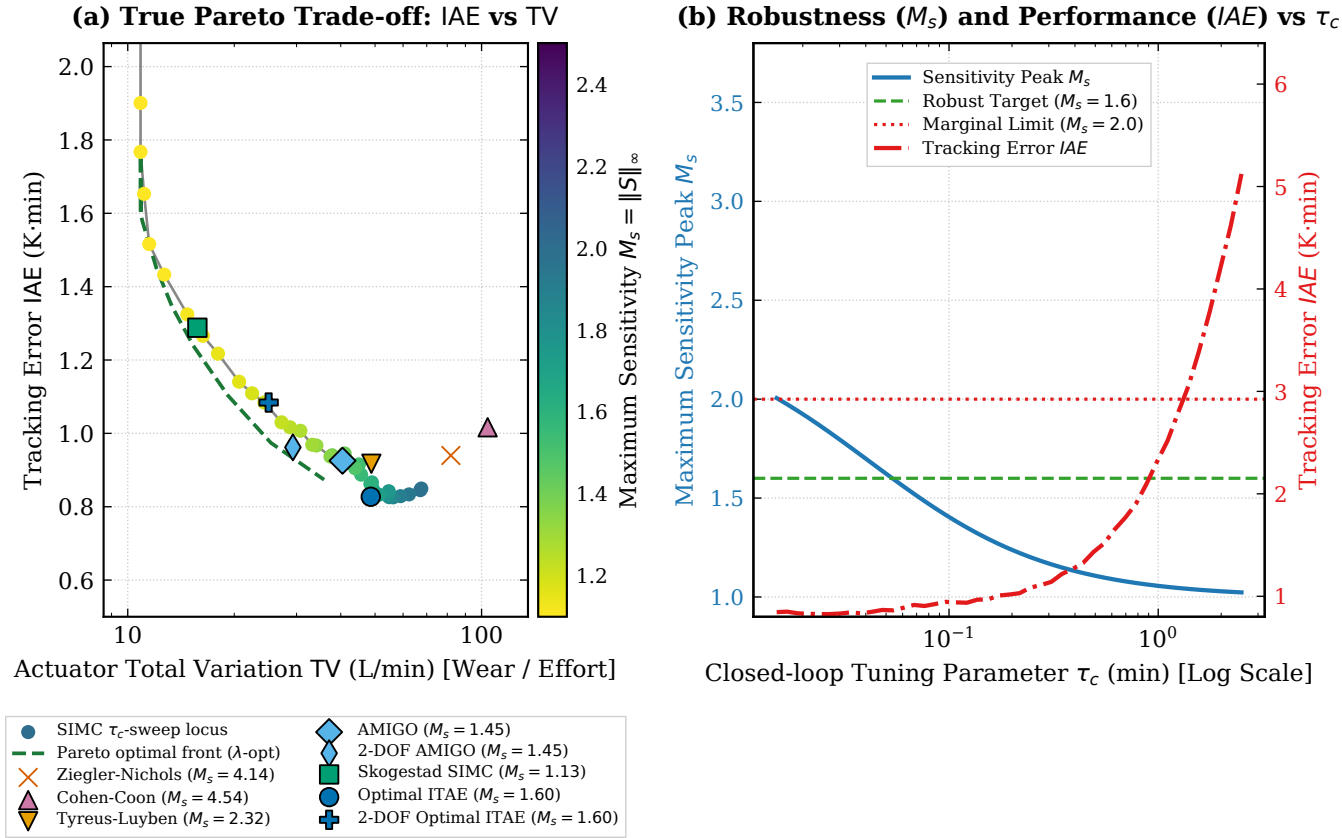
(a) True Pareto Trade-off: IAE vs TV
Tracking Error IAE (K·min)
Actuator Total Variation TV (L/min) [Wear / Effort]
Maximum Sensitivity $M_s = \|S\|_\infty$
(b) Robustness (M_s) and Performance (IAE) vs τ_c
Sensitivity Peak M_s
Robust Target ($M_s = 1.6$)
Marginal Limit ($M_s = 2.0$)
Tracking Error IAE
Maximum Sensitivity Peak M_s
Tracking Error IAE (K·min)
Closed-loop Tuning Parameter τ_c (min) [Log Scale]
SIMC τ_c-sweep locus
Pareto optimal front (λ-opt)
Ziegler-Nichols ($M_s = 4.14$)
Cohen-Coon ($M_s = 4.54$)
Tyreus-Luyben ($M_s = 2.32$)
AMIGO ($M_s = 1.45$)
2-DOF AMIGO ($M_s = 1.45$)
Skogestad SIMC ($M_s = 1.13$)
Optimal ITAE ($M_s = 1.60$)
2-DOF Optimal ITAE ($M_s = 1.60$)